Honeywell

Honeywell Versatilis™ Transmitter

Release 100

Equipment Health Monitoring (EHM)

Installation and User's Guide

34-VT-25-01
February 2023

DISCLAIMER

This document contains Honeywell proprietary information. Information contained herein is to be used solely for the purpose submitted, and no part of this document or its contents shall be reproduced, published, or disclosed to a third party without the express permission of Honeywell International Inc.

While this information is presented in good faith and believed to be accurate, Honeywell disclaims the implied warranties of merchantability and fitness for a purpose and makes no express warranties except as may be stated in its written agreement with and for its customer.

In no event is Honeywell liable to anyone for any direct, special, or consequential damages. The information and specifications in this document are subject to change without notice.

CONTENTS

CHAPTER

1

ABOUT THIS GUIDE

This guide provides information to assist you in configuration and user management of the Honeywell Experion EHM.

Revision history

Revision	Date	Description
A	February 2023	The initial release of the document for R100.

Related documents

Document Name	Document Number
Honeywell Versatilis Connect App User's Guide	34-VT-25-03
Honeywell Versatilis Transmitter Technical Specification	34-VT-03-01
Experion EHM Configuration and User's Guide	34-VT-25-04

Terms and abbreviations

Terms	Definitions
ATEX	Appareils destinés à être utilisés en Atmosphères Explosives
BLE	Bluetooth® Low Energy
CCoE	Chief Controller of Explosives
CAPEX	Capital Expenditures

Terms	Definitions
iOS	iPhone Operating System
IIoT	Industrial Internet of Things
LoRa	"Long Range" Radio Communication Technique
LPWA	"Low Power, Wide Area" networking protocol
MPS	Molecular Property Spectrometer
OPEX	Operational Expenditure
UI	User Interface
UOM	Unit of Measure

CHAPTER

2

INSTRUCTIONS AND SAFETY MEASURES

Precautions

The following precautions must be exercised to use the Honeywell Versatilis Transmitter safely and effectively:

- Honeywell will not provide any guarantee, if the Honeywell Versatilis Transmitter is disassembled.
- The battery may present a potential electrostatic ignition hazard when dissembled.
- Improper use may lead to battery fluid leakage, excessive heat, ignition, or explosion.
- Honeywell will not be liable for any hazard that might be caused due to negligence in handling the Honeywell Versatilis Transmitter.
- Care should be taken to protect this Honeywell Versatilis Transmitter from impact or abrasion if located in a Zone 0/ Class I Div 1 environment.
- It is the responsibility of the end user to verify that the Honeywell Versatilis Transmitter has the necessary approvals required for the intended area of use.
- Verify that the operating environment of the Transmitter is consistent with the appropriate hazardous location’s certification.

Hazardous locations

Honeywell Versatilis Transmitter is available with IECEx, ATEX, UKCA Ex, North America Class I Div I & CCoE approvals.

For more details, see certifications and the same information is also available in *Honeywell Versatilis Transmitter Technical Specification* document.

Best practices

Table 2-1: Best practices - DOs

DOs	
✓	Ensure there is adequate space to access the Honeywell Versatilis Transmitter before selecting an installation position.
✓	It is recommended to install the Honeywell Versatilis Transmitter vertically perpendicular to the rotating shaft.
✓	Pay special attention in case of necessity to install the Honeywell Versatilis Transmitter horizontally on the target machine.
✓	While using a magnetic mounting adapter, the vibration measurement frequency band drops. Ensure the target surface is free from greasy, corrosion, abrasion, and uneven surfaces so that the magnetic adapter attaches firmly to the target, thereby improving the measurement of frequency response.
✓	While using a magnetic mounting adapter, the surface temperature accuracy may vary, if the surface is not clean enough. Ensure the target surface is free from greasy, corrosion, abrasion, and uneven surfaces so that the magnetic adapter attaches firmly to the target, thereby improving the accuracy of surface temperature.
✓	Ensure there are no damages, pigments, dents, or contortion to the surface of the Honeywell Versatilis Transmitter base. Any such deformations or pigments may affect performance and measurement accuracy.
✓	Perform suitable adjustments to align the x-axis, y-axis, and z-axis to the desired orientation on the target machine.
✓	Dispose the transmitter and battery according to the local laws and regulations.
✓	Installation in an explosive environment must be under the appropriate local, national, and international standards,

	DOs
	codes, and practices.

Table 2-2: Best practices - DON'Ts

	DON'Ts
	Do not remove the Honeywell Versatilis Transmitter enclosure without written consent from Honeywell. The Honeywell Versatilis Transmitter is packaged with battery and sensitive electronics, that may be damaged without proper care. Pay attention to don’ts when the Honeywell Versatilis Transmitter enclosure is removed: • Do not short-circuit. • Do not disassemble or change • Do not expose to heat or fire.
	Avoid using bare hands while installing the magnetic adapter on the target machine.

CHAPTER 3

HONEYWELL VERSATILIS TRANSMITTER OVERVIEW

Honeywell Versatilis Transmitter is a multi-variant sensing platform based on the latest LoRaWAN® protocol communication technology. It's inherently low power compact design coupled with quick and easy installation and commissioning help customers to deploy them at scale with the lowest CAPEX and negligible OPEX.

Key features

The key features of Honeywell Versatilis Transmitter are:

Table 3-1: Key features of Honeywell Versatilis Transmitter

LoRa	Based on the latest LoRaWAN® protocol communication technology for large area coverage.
	Built-in battery lasts for years.
	Quick and easy installation & commissioning.
	Robust, and intrinsically safe. Built-in environmental compensation.
	Access real time and historical data visualization.
	Configurable sensor parameters, and data update frequency rate.
	Multiple mounting options available.

Illustrations and dimensions

The physical dimensions of the Honeywell Versatilis Transmitter are shown below:

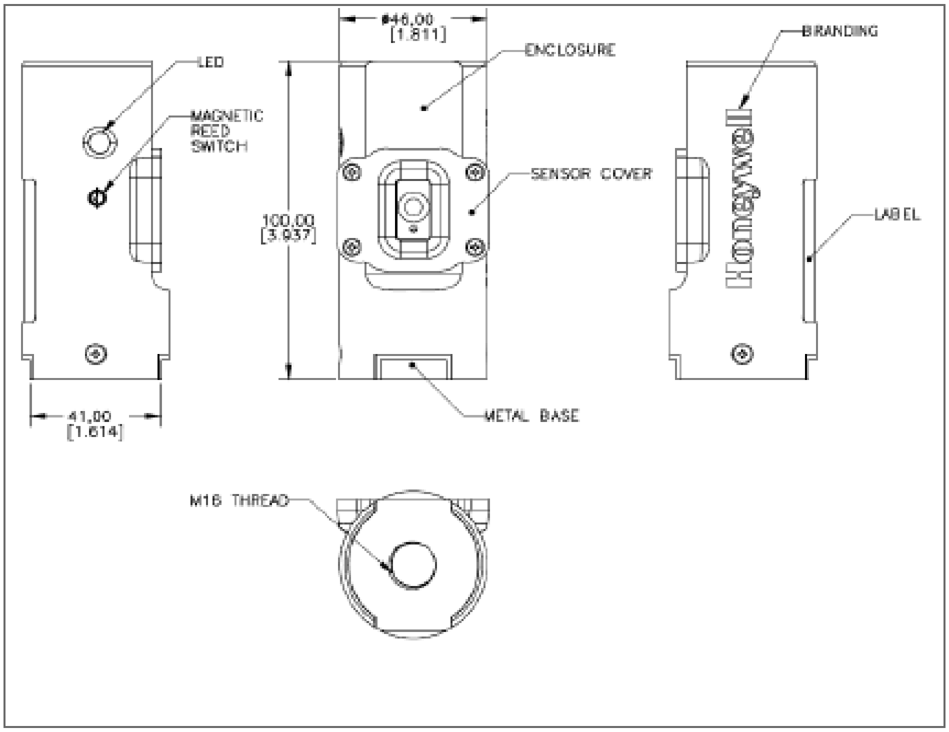


Figure 3-1: Honeywell Versatilis Transmitter dimensions

CHAPTER

4

SPECIFICATIONS

Hardware Specifications

The hardware specifications of Honeywell Versatilis Transmitter are:

Table 4-1: Honeywell Versatilis Transmitter - Hardware Specifications

Parameters	Description
Measurement parameters	Surface Temperature, Ambient Temperature, Ambient Pressure, Humidity, Vibration, and Audio Acoustics.
Communication	LoRaWAN® Class-A. For information on Honeywell Versatilis Transmitter LoRaWAN® Frequency and Channel details, see *Honeywell Versatilis Transmitter Technical Specification.*
	2.4 GHz Bluetooth Low Energy
Battery life	5 years (with 5min sensor measurement interval & 30min LoRa update)
Battery voltage	3.6V DC
Honeywell Versatilis Transmitter LED status	For more information, see LED states.
Operating temperature	-40°C to +80°C (-40°F to +176°F)
Dimensions	W: 46 mm (1.81 Inches) x H: 100 mm (3.93 Inches)
Weight	180 grams
Mounting adapters	Magnetic, Screw, and Adhesive mount adapters.

Parameters	Description
	For more details, see Mounting options.

Environmental conditions

The following table provides the operating conditions of Honeywell Versatilis Transmitter:

Table 4-2: Environmental conditions of Honeywell Versatilis Transmitter

Information	Value
Tamb range	-40°C to +80°C (-40°F to +176°F)
Relative Humidity	5% to 100%
Usage	Indoor and Outdoor
*NEMA Type 4X rating approval is in progress. This will enable Outdoor installations for North America region	

For more information, see *Honeywell Versatilis Transmitter Technical Specification* sheet.

Product label:

The following figure shows the Honeywell Versatilis Transmitter product label. The contents of the label will be printed on the product.

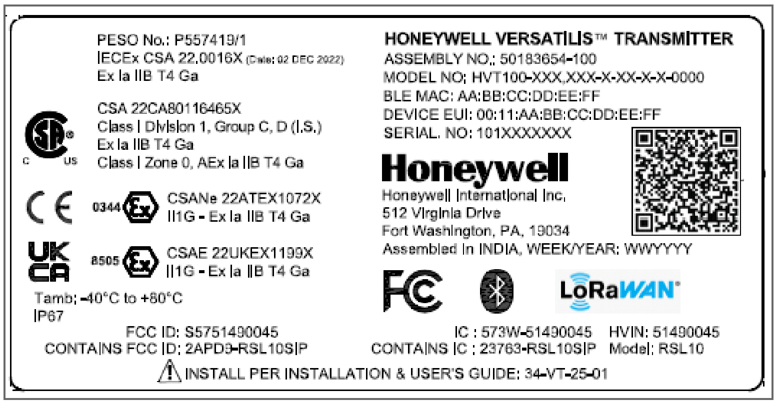


Figure 4-1: Honeywell Versatilis Transmitter label

CHAPTER

5 SETTTING UP THE HONEYWELL VERSATILIS TRANSMITTER

Unpacking the contents

The Honeywell Versatilis Transmitter comes in a cardboard shipping container as shown in the following figure.

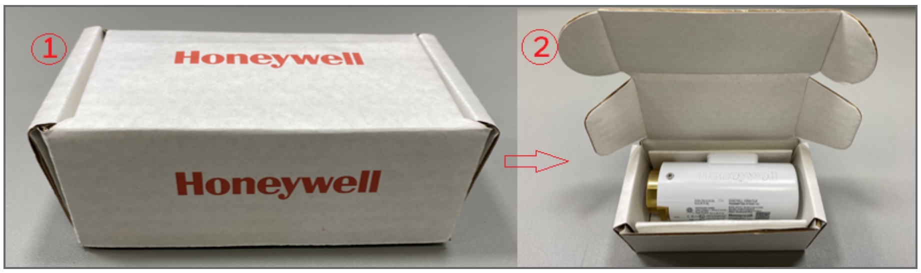


Figure 5-1: Unpacking the Honeywell Versatilis Transmitter[1]

> **NOTE:** After unpacking the Honeywell Versatilis Transmitter, it is recommended to check for any visible damage to the filter on the front of the Transmitter. If the filter is damaged then the sensor will not function as intended. In such cases of damaged filter, contact HPS Technical Support through your local Customer Contact Center.

[1]The packaging design and information included in this document are related to the proposed prototype, and is yet to be finalized.

The magnetic and adhesive mount adapters are supplied as kits along with Honeywell Versatilis Transmitter, if chosen the same while ordering.

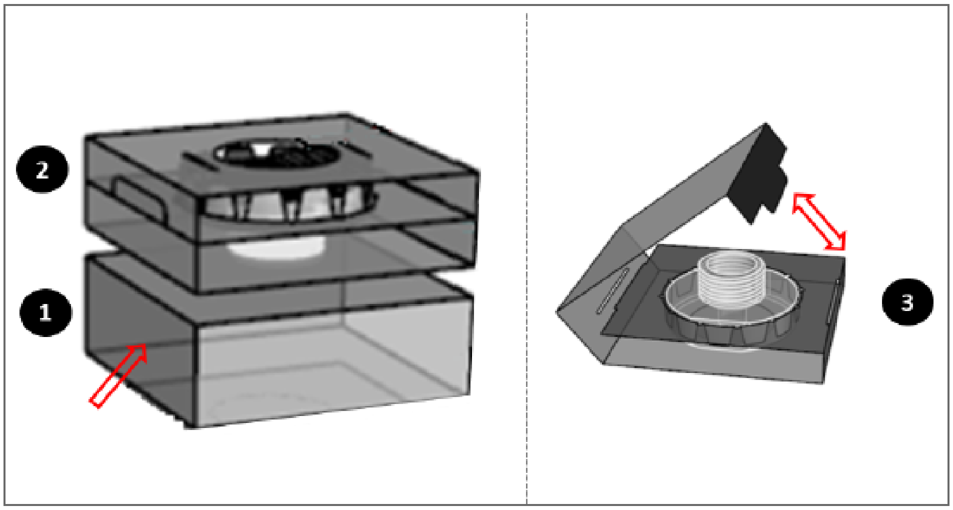


Figure 5-2: Unpacking mounting adapters

The following table provides the contents of the shipping package, while ordering the Honeywell Versatilis Transmitter with each adapter variants:

For Honeywell Versatilis Transmitter ordered with adapter variants	Contents in the package	Accessories/ Tools (To be handy with users)
Screw Mount	Honeywell Versatilis Transmitter fitted with Screw Mount (Default option)	• M6 Screw and M6 Nut (based on target machine requirements). • Allen key for M6 socket head cap screw, size: 5mm. • The spanner recommended for the Signal Scout's base is 41mm, and for the Adapter is

For Honeywell Versatilis Transmitter ordered with adapter variants	Contents in the package	Accessories/ Tools (To be handy with users)
		32mm.
Magnetic Mount	Honeywell Versatilis Transmitter, Magnetic mounting adapter	The spanner recommended for the Honeywell Versatilis Transmitter's base is 41mm, and for the Adapter is 32mm.
Adhesive Mount	Honeywell Versatilis Transmitter, Adhesive mounting adapter	The spanner recommended for the Honeywell Versatilis Transmitter's base is 41mm, and for the Adapter is 32mm.

ATTENTION: For disposing off the recyclable Transmitter and packaging materials, it is recommended to first remove the battery from the Honeywell Versatilis Transmitter. Then dispose it separately as per the manufacturer's recommendations, in compliance with the concerned laid down regulations.

Mounting Honeywell Versatilis Transmitter

The Honeywell Versatilis Transmitter offers multiple mounting options such as Magnetic Mount, Adhesive Mount, and Screw Mount to suit the mounting surface of the target machine to ensure good bonding and accurate measurement.

NOTE: Users need to select the suitable mounting adapter while placing an order.

The following table includes the list of recommended tools that are required for installation/ replacement scenarios:

Tool	Size
Spanner (for firmly holding the base of Honeywell Versatilis Transmitter while tightening the mounting adapter to it).	41 mm
Spanner (for tightening the mounting adapter to the base of the Honeywell Versatilis Transmitter)	32 mm
Allen Key and Spanner (for firmly holding the M6 socket head cap screw while tightening with M6 nut respectively)	Allen key size: 5mm; and its respective spanner size: 10mm.

Magnetic Mounting

Follow the below procedure to install the magnetic mounting on the target machine:

> **ATTENTION:** Do not use bare hands while installation as the magnet is powerful and can pinch the skin/ fingers if not handled properly.

1. Screw-in the magnetic mount adapter into the threaded hole provided on the base of the Honeywell Versatilis Transmitter.
2. Firmly hold the base of the Honeywell Versatilis Transmitter using a spanner (of size 41mm), and tighten the adapter to the base using another spanner (of size 32mm).
   Ensure a Torque of 3.5 to 4 Nm is applied for tightening.
3. Attach the Honeywell Versatilis Transmitter fitted with a magnetic mount adapter to the target machine, with the magnetic pull force.

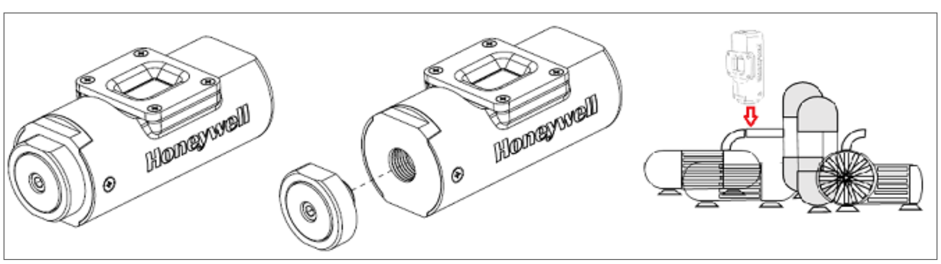


Figure 5-3: Magnetic mounting

Adhesive Mounting

Perform the below instructions for adhesive mounting on the target structure:

Prerequisite: Remove oil, moisture, and dirt from the intended mounting surface of the target structure on which the Honeywell Versatilis Transmitter will be mounted. If the dirt is strong, remove it with isopropyl alcohol.

> **ATTENTION:** Use the adhesive mounting adapter preferably on a flat surface. As there is a potential risk of falling down of the Honeywell Versatilis Transmitter if it is mounted on uneven, rough, or curved surfaces, due to lack of sufficient bonding area.

1. Screw-in the adhesive mount adapter into the threaded hole provided on the base of the Honeywell Versatilis Transmitter.
2. Firmly hold the base of the Honeywell Versatilis Transmitter using a spanner (of size 41mm), and tighten the adapter to the base using another spanner (of size 32mm). Ensure a Torque of 3.5 to 4 Nm is applied for tightening.
3. Remove the protective film from adhesive face of the adapter.
4. Stick the Honeywell Versatilis Transmitter fitted with an adhesive adapter onto the target machine. Apply an adequate pressure on the Honeywell Versatilis Transmitter after it is mounted, to ensure proper bonding of the pressure-sensitive adhesive with the mounting surface.

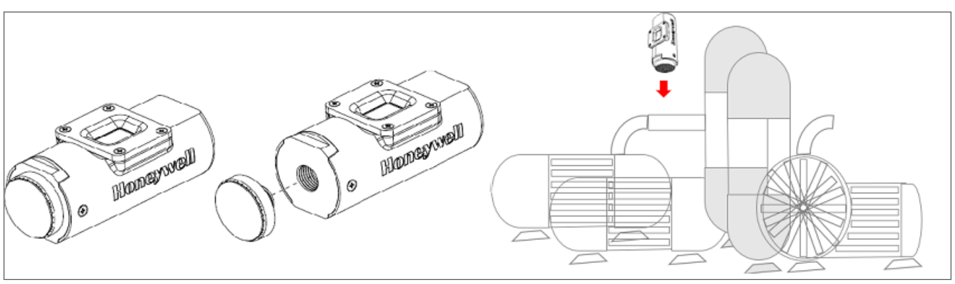


Figure 5-4: Adhesive mounting

Screw Mounting

Perform the below instruction for screw mounting on the target structure:

1. Insert the M6 socket head cap screw into the hole provided on the screw-mount adapter (where the head of the screw sits inside the adapter and the shank protrudes outwards).
2. Insert the protruding M6 socket head cap screw (with adapter) into the hole provided on the target structure/bracket, and then secure the adapter with M6 nut (on the other side of structure/bracket) using Allen key and spanner (of 10mm). Ensure a Torque of 16 N-m or 140 in-lbs is applied for tightening. Or, If there is an existing M6 tapped hole on the target structure, then you just need to insert the protruding M6 socket head cap screw (with adapter) into that hole provided on the target structure/bracket, and tighten with Allen key (of required size).
3. Fit the Honeywell Versatilis Transmitter onto the secured adapter. Firmly hold the secured adapter using a spanner (of size 32mm), and tighten the base of the Honeywell Versatilis Transmitter to the adapter using another spanner (of size 41mm). Ensure a Torque of 3.5 to 4 Nm is applied for tightening.

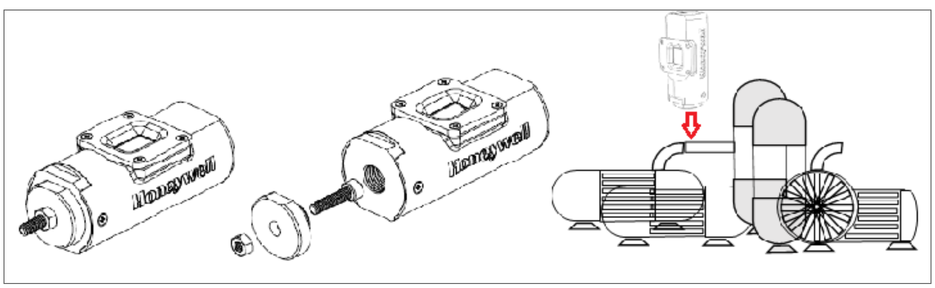


Figure 5-5: Screw mounting

CHAPTER

6 CONFIGURATION

The following figure and table provide information about the complete solution architecture designed and implemented for Honeywell Versatilis Transmitter. This information helps the user to understand the various process involved, right from the configuration of the Honeywell Versatilis Transmitter to access the analytic solutions.

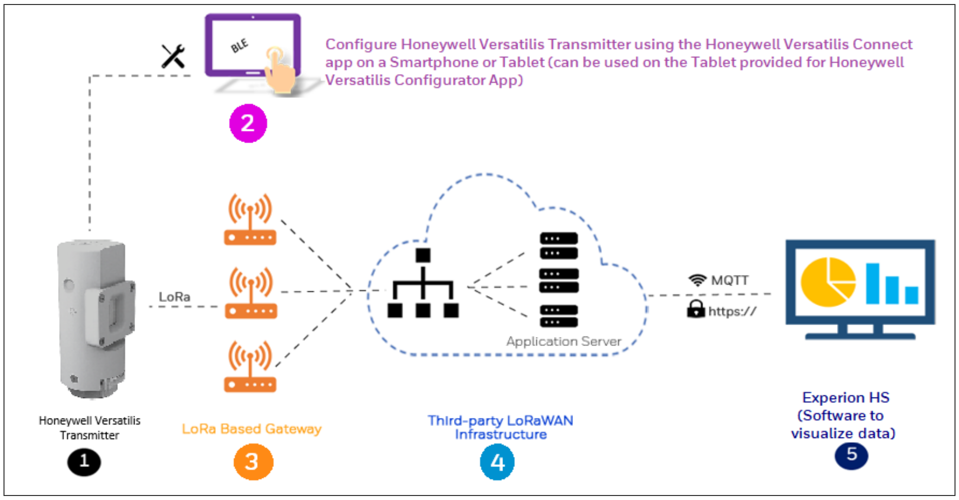


Figure 6-1: Architecture of Honeywell Versatilis Transmitter

Table 6-1: Description of architecture flow

Item	Integration	Description
1	Honeywell Versatilis Transmitter	The Honeywell Versatilis Transmitter measures six parameters on the target machine, which are as follows: • Surface temperature • Ambient Humidity • Ambient temperature • Ambient pressure • Vibration • Audio acoustics

Item	Integration	Description
2	Honeywell Versatilis Connect App	The Honeywell Versatilis Connect App enables user to connect to the Honeywell Versatilis Transmitter through bluetooth using a tablet, or smartphone. It helps user to configure the end Transmitter's sensor parameters, view live data, update firmware, etc.
3	LoRa Gateway	The third-party LoRa based gateways acts as a medium to push the sensor data from the Honeywell Versatilis Transmitter to the LoRaWAN infrastructure in a secured way
4	LoRaWAN Infrastructure	.The third-party LoRaWAN Infrastructure applies the payload formatter to decrypt the incoming data from the LoRa gateways, and securely transfer it through the MQTT protocol to the Experion EHM.
5	Honeywell Experion EHM	The Honeywell Experion EHM software provides the platform to visualize the transmitted sensor data and provide useful insights to monitor and track the health of the target machines.

Installation of the Honeywell Versatilis Connect app

The Honeywell Versatilis Connect app provides flexibility to install on Smartphone or Tablet, that supports Android or Microsoft Windows platform. Users can also use the Tablet (if any) provided with Honeywell Versatilis Configurator App to install and run the Honeywell Versatilis Connect app with ease.

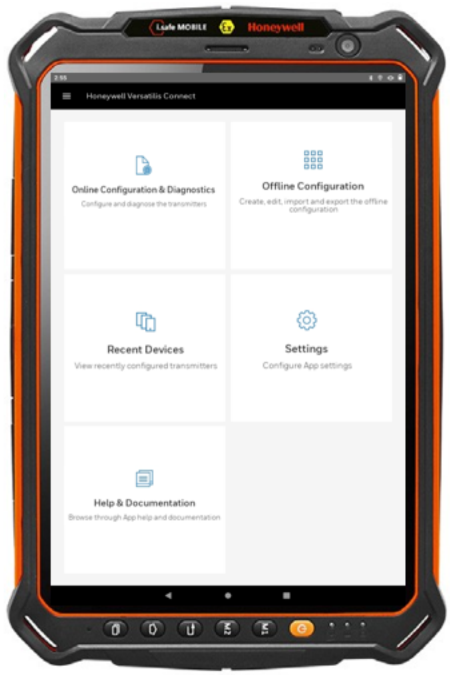


Figure 6-2: Honeywell Versatilis Connect app installed on "Honeywell Versatilis Configurator Tablet"

Prerequisites:

Table 6-2: Prerequisites of Honeywell Versatilis Connect app

Tablet / Smartphone Specifications	Windows	Android
Operating System	Windows 10 or higher versions	Android 10 or higher versions
Process and Speed	64-bit, 1.6GHz or faster	ARM V7 or V8, 1.6GHz or faster
RAM	Minimum: 8GB	Minimum: 4GB Recommended: 8GB
Storage space	Higher than 64GB is recommended	Higher than 64GB is recommended

To download and install the Honeywell Versatilis Connect app, follow the procedure described below:

Table 6-3: Installation procedure of Honeywell Versatilis Connect app

For Windows Platform	For Android Platform
1. Open the **Microsoft Store** app, and search for Honeywell Versatilis Connect. 2. Tap **Get** to install. 3. Tap **Open**.	1. Open the **Google Play Store** app, and search for Honeywell Versatilis Connect. 2. Tap **Install**. 3. Tap **Open**. In the permissions dialog, tap **Allow**.

When the user launches the Honeywell Versatilis Connect App for the first time, the app gives a tour of its overall features.

For more information on how to configure Honeywell Versatilis Transmitter, see *Honeywell Versatilis Connect App User's Guide.*

Resetting passcode for the Honeywell Versatilis Connect app:

In case you forgot your changed passcode, you can reset your current passcode to the default passcode using the reed switch provided on the Honeywell Versatilis Transmitter.

Perform the below instructions to reset your current passcode to the default passcode:

> **NOTE:** This activity must be performed within the predefined time period of 15 seconds.

1. Bring a magnet closer to the reed switch location on the Honeywell Versatilis Transmitter. The LED blinks red light whenever you bring the magnet closer to the reed switch.
2. Repeat step 1. A red light blinks at the reed switch location for the second time.
   Repeat the second swipe within 15 seconds.
3. The LED blinks green light twice after the predefined time period of 15 seconds, returning to the normal state, and thereby indicating a successful reset of the current passcode to the default one.

> **NOTE:** After successful reset to the default passcode, you can use the Honeywell Versatilis Connect app to further change the default passcode as required. For more information, see *Honeywell Versatilis Connect App user's guide, 34-VT-25-03*.

Configure LoRa Gateway

LoRa Gateways acts as communication bridge between Honeywell Versatilis Transmitter and LoRaWAN application.

> **NOTE:** There are many LoRa based Gateway manufacturers in the market. Choose the LoRa gateway best suited for your needs. For more information on how to set up the LoRa Gateway, refer the respective manufacturer's product documentation.

Following are some common parameters for your reference that needs to be configured for LoRa gateway:

Functions	Key Parameters
Gateway Information	Displays information such as Model Number, Serial Mumber, IMEI, Frequency Band, Gateway EUI, LAN, WAN, Ethernet, and etc. User can verify these information by checking the same on the sticker provided on the panel of the gateway.
LoRaWAN/ Network Settings	Below information needs to be configured: • LoRa Mode Selection: Packet Forwarder Make sure the status of Packet Forwarder shows as "Running". • Network selection: E.g. Radio Bridge ChirpStack, The Things Network, Senet, Loriot, etc. • Frequency Plan selection
Connection Configuration	The available connection options for connecting gateway to the LoRaWAN are Ethernet, Cellular,

Functions	Key Parameters
	or Wi-Fi. These options for configuring connections are based on the gateway models as offered by the manufacturer.
After gateway set up	Make sure the gateway is powered ON. The status should show as Connected in the LoRaWAN service provider application.

Configure LoRa Network (LoRaWAN)

LoRaWAN is a Low Power, Wide Area (LPWA) networking protocol based on LoRa radio modulation technique. Here, it wirelessly connects battery operated Honeywell Versatilis Transmitter over MQTT and manages communication between Honeywell Versatilis Transmitter and network gateways.

NOTE: There are many LoRaWAN service providers in the market. Choose the LoRaWAN service best suited to your needs. For more information on how to configure Honeywell Versatilis Transmitter and LoRa Gateway in the LoRaWAN service provider application, refer to the respective service provider's product documentation.

LoRa Network server (LoRaWAN) Setup

Prerequisite: The following is the list of actions a user needs to do on their own to set up the LoRa network server on-premise:

- A physical PC with specifications.
- A licensed VMWare Workstation Pro 17 installed on the PC.
- A LoRa network service or Enterprise version from the LoRaWAN service provider of your choice.
- A VM created with Ubuntu 22.04 or later OS (Linux).
- Network server configuration of the LoRaWAN service provider on Ubuntu. For more information on how to configure the LoRa network server (LoRaWAN), refer to the corresponding user documentation on the portal of your LoRaWAN service provider.

After the LoRa network server is configured, to create gateways, device profiles, applications and add end devices using your LoRaWAN service provider's application UI. Refer the following table for some key configuration fields that are used by popular LoRaWAN service providers.

Table 6-4: LoRaWAN Application Configuration - Important Fields

Functions	Key Parameters
Adding Gateway	Below parameters needs to be configured: • Specify a suitable name for the Gateway being added. • Specify a brief description about the Gateway being added. • Specify a unique ID for identifying the Gateway. • Select the required Network-server, and Service-profile. • Enable the Gateway discovery mode.
Creating Device-profile	• Specify a suitable name for the Device-profile being created. • Select the required Network-server. • Specify the supported "LoRaWAN MAC version" for the device as "1.0.4". • Specify the supported "LoRaWAN Regional Parameters revision" for the device as "RP002-1.0.2". • Select the intended network activation method, i.e. ABP/OTAA. • Specify the "RX1 delay" as "5" seconds. • Provide the decryption key provided by Honeywell to decode data coming from the device through Gateway. For information on LoRaWAN MAC version, LoRaWAN Regional parameters version, Frequency plan, or RX1 delay, see *Honeywell versatilis Transmitter Technical Specification* document.

Functions	Key Parameters
Creating an Application	• Specify a suitable name for the Application being created. • Specify a brief description about the Application being created. • Select the required Service-profile.
Adding a Device to the Application	• Specify a suitable name for the Device being added to the intended Application. • Specify a brief description about the Device being added to the intended Application. • Specify the Device EUI as printed on the enclosure of the Signal Scout, and can also be seen on the **About Device** page of the Honeywell Versatilis Connect app. • Select the intended Device-profile you want to apply for the Device being added. Provide the activation keys for ABP/OTAA based on your network method selection while adding the Gateway. For example, the following are the keys to specified for the OTAA method: • LoRa Device EUI: The unique ID as provided on the enclosure of the Honeywell Versatilis Transmitter, and can also be seen on the About Device page of the Honeywell Versatilis Connect app. • LoRa Application EUI: Specify the 16-digits hexadecimal to identify the join server. This value can be manually specified or generated from the LoRaWAN service provider's application.

Functions	Key Parameters
	• LoRa Application key: Specify the 32-digits hexadecimal to identify the application server. This value can be manually specified or generated from the UI. **NOTE:** Make sure to specify the same activation mode and its respective keys values in both the Honeywell Versatilis Connect app and LoRaWAN service application UI to accomplish successful network mapping. For more information on how to configure Honeywell Versatilis Transmitter with ABP or OTAA method of LoRaWAN in the Honeywell Versatilis Connect app, see the *Honeywell Versatilis Connect App User's Guide*
After all the configuration are done and Gateway is successfully added to the LoRaWAN service provider's application.	Make sure the gateway is powered ON. The Gateway status should show as **Connected** in the LoRaWAN service provider's application.

The following is the flow diagram of the LoRa Network configuration process:

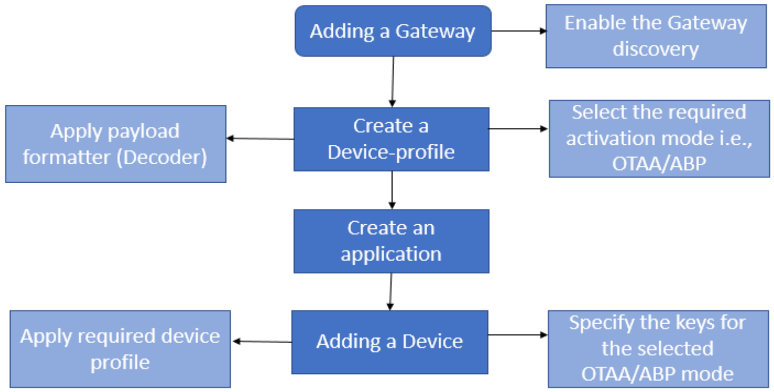


Figure 6-3: LoRa network configuration process - Flow chart

Installation of Honeywell Experion EHM

The Honeywell Experion EHM software provides the platform to visualize the transmitted sensors data and provide useful trends to monitor and track the health of the target machines.

Prerequisite: The following is the list of actions a user needs to take care of on their own to set up the Experion EHM software:

- A on-prem PC with:

Parameters	Description
Operating system	Windows 10 Enterprise 2019 LTSC
Storage space	500 GB SSD/HDD
RAM	32GB
Processor	Single or multiple Intel 2.5GHz, Octa Core or greater
Networking	1GBPS or 100 MBPS Ethernet

- A licensed VMWare Workstation Pro 17 installed on the PC.

Honeywell Experion EHM Setup

Perform the below instructions to access the VM created for Experion EHM:

1. Insert the Pendrive (supplied along with the Honeywell Versatilis Transmitter) into your local machine.
2. Copy the .TAR file from your Pendrive to your local machine.
3. On your local machine, right-click the **TAR file > Extract**.
4. A pop-up window prompts you to specify the encryption key. Specify the encryption key provided to you by Honeywell.

   After validating the specified encryption key, the extraction process of the .TAR file begins. After successful extraction, the .TAR file opens up a new instance of VM including the pre-installed Experion EHM setup. The required licensing model auto-applies as chosen while initiating an order from the factory.

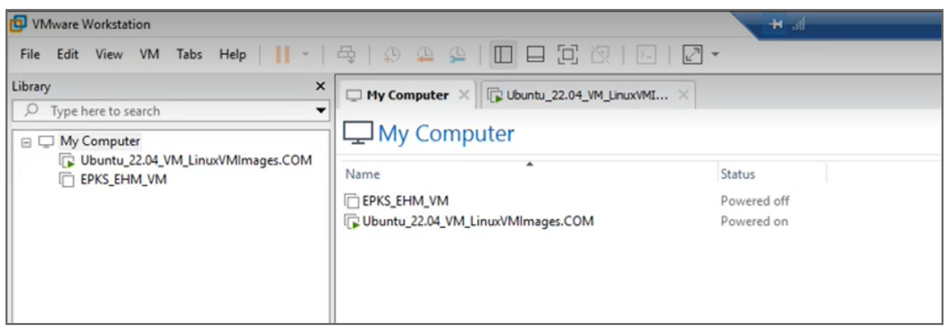


Figure 6-4: VM created for Experion EHM

For more information on how to configure and use Experion EHM Software, see *Configuration and User's Guide for Experion EHM*.

CHAPTER

7 SECURITY

The security mechanisms implementation rely on the well tested and cryptographic algorithms which are analyzed by the cryptographic community, NIST approved and widely adopted as a best security for constrained nodes and networks.

Security features

The secure features of the Honeywell Versatilis Transmitter are:

- Secure firmware update.
- Secure end-to-end Bluetooth and LoRa (i.e., secure data communication by application payload and pairing).
- Authentication on the BLE security using passcode.
- Data protection, Data integrity and Confidentiality protection.
- The security supported by standard BLE and LoRaWAN protocols is well implemented.
- Supports LoRaWAN® Class-A security using OTAA/ ABP activation mode.

Physical security

Keys are persistently stored in the Honeywell Versatilis Transmitter and their protection depends on the Honeywell Versatilis Transmitter's physical security.

For information on support and reporting, see Notices.

CHAPTER

8

MAINTENANCE

Replacement of Honeywell Versatilis Transmitter

The modularized design of the Honeywell Versatilis Transmitter allows the user to disassemble from current target machine and to mount on other target machine with ease, or to replace the currently fitted mounting adapter with some other adapter to suit change in mounting surface.

Perform the below instructions to replace the Honeywell Versatilis Transmitter:

1. Detach the Honeywell Versatilis Transmitter from the target structure:
   - In case of screw-mount: Remove the M6 nut securing the Honeywell Versatilis Transmitter onto the target structure.
   - In case of magnetic mount: Remove the Honeywell Versatilis Transmitter from the target structure manually.
   - In case of adhesive mount: Cut through the adhesive layer (sticking the adapter to the target structure) using a blade.
2. Firmly hold the base of the Honeywell Versatilis Transmitter using a spanner (of size 41 mm), and unscrew the adapter from the base of the Honeywell Versatilis Transmitter using another spanner (of size 32 mm) till it's completely disassembled.
3. Install the Honeywell Versatilis Transmitter with the required mounting adapter. For more information on mounting procedure of Honeywell Versatilis Transmitter with various adapter types, see Mounting options.

Firmware update

The Honeywell Versatilis Connect app has the provision to update to the latest firmware available for the Honeywell Versatilis Transmitter. For more information, see the *Honeywell Versatilis Connect App User's Guide.*

Device logs

User has the provision to download the log files and save it to the local drive using Honeywell Versatilis Connect app.

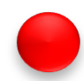
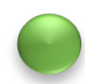
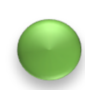
For more information on how to download the Honeywell Versatilis Transmitter's logs using Honeywell Versatilis Connect app, see the *Honeywell Versatilis Connect app User Guide.*

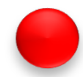
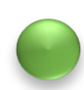
CHAPTER

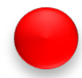
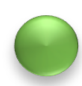
9 LED STATES

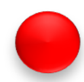
The following table provides various states of Signal Scout's LEDs and their associated status based on different scenarios.

Table 9-1: LEDs States

Probable Scenarios	Honeywell Versatilis Transmitter status	LED
On inserting the battery at the factory.	Power ON	 (1. Blinks once)  (2. Blinks twice) After 19 seconds:  (3. Blinks thrice)
	Power ON failure	No visual indication on LED
Connecting to the Honeywell Versatilis Transmitterusing Bluetooth scan in Honeywell Versatilis Connect app.	Successful pairing	 (Blinks once)

Probable Scenarios	Honeywell Versatilis Transmitter status	LED
User selects the Honeywell Versatilis Transmitter from the list of "Available Devices" displayed on the Honeywell Versatilis Connect app.	Unsuccessful pairing	 (Blinks once)
	Not recognizing	No visual indication on LED.
Connecting to the Honeywell Versatilis Transmitter using QR code scan in the Honeywell Versatilis Connect app.	Successful pairing	 (Blinks once)
Scanning QR code imprinted on the Honeywell Versatilis Transmitter, flashes the Transmitter's summary in the Honeywell Versatilis Connect app.	Unsuccessful pairing	 (Blinks once)
	Not recognizing	No visual indication on LED.
User activates the Honeywell Versatilis Transmitter to start measuring parameters, or pushing configurations to the Transmitter as required from the Honeywell Versatilis Connect app.	During configuration	
On successful configuration of the Honeywell Versatilis Transmitter.	Configuration successful	 (Blinks thrice)

Probable Scenarios	Honeywell Versatilis Transmitter status	LED
Configuring the Honeywell Versatilis Transmitter using "Offline Template".	Configuration failure	 (Blinks thrice)
Activate/ Deactivate through Connect App.	On activation	 (Blinks thrice)
	Deactivate	No visual indication on LED.
Passcode reset using reed switch	Successful	 (Blinks twice)
	Unsuccessful	No visual indication on LED.
Firmware update	While downloading/ updating	 (Blinks for every 10 seconds)
	Successful updating, and restarting	 (1. Blinks

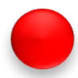
Probable Scenarios	Honeywell Versatilis Transmitter status	LED
		once)  (2. Blinks twice) After 19 seconds:  (3. Blinks thrice)
	Unsuccessful	

CHAPTER 10 TROUBLESHOOTING

The following table provides various troubleshooting scenarios in the case of error or unexpected behavior, and their corresponding troubleshooting tips:

Table 10-1: Troubleshooting Information

Scenarios	Honeywell Versatilis Transmitter Status	LED	On Screen	Troubleshooting Tips
User selects the Honeywell Versatilis Transmitter from the list displayed on the Honeywell Versatilis Connect.	Unsuccessful pairing.	(Blinks once.)	A pop-up window prompts to try again.	• Retry pairing. • Verify the passcode specified for Honeywell Versatilis Connect authentication.
	Not recognizing.	No visual indication on LED.		Ensure that the device is in within the BLE range.
Scanning QR code imprinted on the Transmitter, flashes the Honeywell Versatilis Transmitter 's summary.	Unsuccessful pairing.	(Blinks once.)	A pop-up window prompts to try again.	• Re-scan the QR code. • Connect manually through BLE app.
	Not recognizing.	No visual indication on LED.	NA	

Scenarios	Honeywell Versatilis Transmitter Status	LED	On Screen	Troubleshooting Tips
Firmware update	Unsuccessful	 (Blinks thrice.)	A pop-up window with error message appears.	• Check the BLE signal strength. Ensure that device is connected to the Connect app. Re-try firmware update. • Ensure the required firmware file is downloaded from authenticated location. • Contact Honeywell TAC team.
Higher vibration alarms during asset start-up in batch operations can trigger a false alarm to Experion EHM solution (this is because of asset	NA	NA	NA	• Ignore the alarm. • Alarm status will get updated with current outstanding alarms if any, during its regular alarm publish rate of 8hrs

Scenarios	Honeywell Versatilis Transmitter Status	LED	On Screen	Troubleshooting Tips
generating higher vibration signatures during start-up conditions).				
Diagnostic fault indication.	Battery low.	No visual indication on LED.	Honeywell Versatilis Connect app shows  with battery indication. screens and Experion EHM.	Honeywell Versatilis Transmitter replacement.
	Sensor interface failure.		Status indicator at sensor level as well as on the Diagnostic page of Honeywell Versatilis Connect and Experion EHM.	• Restart Honeywell Versatilis Transmitter. • Replace Honeywell Versatilis Transmitter.
	LoRa communication status.		Communication fault indication on the Diagnostic page of Honeywell Versatilis Connect and Experion EHM.	• Install the Gateway within the reachable range of the device as per the LoRaWAN standard. • Ensure the device is

Scenarios	Honeywell Versatilis Transmitter Status	LED	On Screen	Troubleshooting Tips
				configured in the LoRaWAN server with valid keys (for ABP/OTAA method). • Ensure the device is configured in Honeywell Versatilis Connect with valid keys (for ABP/OTAA method). • Ensure the keys specified in Honeywell Versatilis Connect and LoRaWAN server are the same. • Ensure valid LoRa reporting interval is set in Honeywell Versatilis Connect. • Restart Honeywell Versatilis Transmitter.

CHAPTER

11

CERTIFICATIONS

Hazardous location certifications

Honeywell Versatilis Transmitter is certified for various Hazardous location standards and requirements.

The below tables gives the summary on the same:

Table 11-1: Hazardous location certifications

Certification	Standards	Approval/ Rating
IECEx	IEC 60079-0/COR1: 2020; Edition 7.0; 2017-12 IEC 60079-11: Edition 6.0; 2011-06	Ex ia IIB T4 Ga Tamb: -40°C to +80°C
CE - ATEX (2014/34/EU)	EN 60079-0: 2018 EN 60079-11: 2012	II 1 G - Ex ia IIB T4 Ga Tamb: -40°C to +80°C
UKCA Ex	EN 60079-0: 2018 EN 60079-11: 2012	II 1 G - Ex ia IIB T4 Ga Tamb: -40°C to +80°C
North America	CAN/CSA C22.2 No. 61010-1-12 + UPD1: 2015, UPD2: 2016, AMD 1-18 CAN/CSA C22.2 No. 60079-0: 19 CAN/CSA-C22.2 No. 60079-11: 14 (R2018) ANSI/UL 61010-1-2018 Third Edition	Class I Division 1, Group C, D Ex ia IIB T4 Ga Class I Zone 0, AEx ia IIB T4 Ga Tamb: -40°C to +80°C

Certification	Standards	Approval/ Rating
	ANSI/UL 913-2019 Eighth Edition ANSI/UL 60079-0-2020 Seventh Edition ANSI/UL 60079-11-2018 Sixth Edition	
CCoE	IS/IEC 60079-0: 2017 IS/IEC 60079-11: 2011	Ex ia IIB T4 Ga Tamb: -40°C to +80°C

Specific conditions of use:

- The nonmetallic enclosure parts of this equipment may become a spark ignition hazard in the presence of static electricity. The enclosure shall be cleaned only with a damp cloth, and the equipment shall be mounted to avoid building static electric charge from non conductive process flow, strong air currents, or other potential charging through friction.
- The aluminum enclosure may be capable of producing incendive sparks when impacted. This equipment must be mounted and/or physically guarded such that it is not subjected to impact or friction.

WARNING: DO NOT REPLACE BATTERY IN EXPLOSIVE ENVIRONMENT.
USE ONLY "ER18505 BATTERY FROM EVE Energy Co., Ltd." or "SB-A01 from Vitzrocell, Co., Ltd".

CE (Conformance to Europe)

Honeywell Versatilis Transmitter is compliant with all the Directives that are applicable as per CE certification requirements.

The below tables gives the summary on the same:

Table 11-2: CE certification requirements

Certification	Standards	Directive/ Regulation
CE	EN 61326-1: 2013 EN 61326-2-3: 2013 EN55011: 2009 + A1: 2010 EN 61000-4-2: 2009 EN 61000-4-3: 2006+A1+A2 EN 61000-4-8: 2010	Electro Magnetic Compatibility (EMC) Directive; 2014/30/EU
CE	ETSI EN 300 328 ETSI EN 300 220-1 V3.1.1 (2017-02) ETSI EN 300 220-2 V3.1.1 (2017-02) ETSI EN 301 489-1: 2019 ETSI EN 301 489-3: 2021 ETSI EN 301 489-17: 2020	Radio Equipment Directive (RED); 2014/53/EU
CE	EN 61010-1: 2010/A1: 2019	Low Voltage Directive (LVD); 2014/35/EU
CE	EN 50581: 2012	Restriction of use of Hazardous Substances (RoHS) in Electrical and Electronic

Certification	Standards	Directive/ Regulation
		equipment; 2011/65/EU; 2017/2102 amendment
CE	EN 50385: 2017	Minimum health and safety requirements regarding the exposure of workers to the risks arising from physical agents (Electromagnetic fields); 2013/35/EU

UKCA (United Kingdom Conformity Assessed)

Honeywell Versatilis Transmitter is compliant with all the Regulations that are applicable as per UKCA certification requirements.

The below tables gives the summary on the same:

Table 11-3: UKCA certification requirements

Certification	Standards	Directive/ Regulation
UKCA	EN 61326-1: 2013 EN 61326-2-3: 2013 EN55011: 2009 + A1: 2010 EN 61000-4-2: 2009 EN 61000-4-3: 2006+A1+A2 EN 61000-4-8: 2010	Electro Magnetic Compatibility (EMC) Regulations 2016
UKCA	ETSI EN 300 328	Radio Equipment Regulations 2017

Certification	Standards	Directive/ Regulation
	ETSI EN 300 220-1 V3.1.1 (2017-02) ETSI EN 300 220-2 V3.1.1 (2017-02) ETSI EN 301 489-1: 2019 ETSI EN 301 489-3: 2021 ETSI EN 301 489-17: 2020	
UKCA	EN 61010-1: 2010/A1: 2019	Electrical Equipment (Safety) Regulations 2016
UKCA	EN 50581: 2012	Restriction of the Use of Certain Hazardous Substances (RoHS) in Electrical and Electronic Equipment Regulations 2012
UKCA	EN 50385: 2017	The Control of Electromagnetic Fields at work Regulations 2016

FCC & IC Certifications

Honeywell Versatilis Transmitter is complaint with all the requirements that are applicable as per FCC & IC certification specifications.

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

The below tables gives the summary on the same:

Table 11-4: FCC and IC certification requirements

Certification	Standards	Approval/ Rating
FCC	47 CFR Part 15 [10-01-20 Edition] ANSI C63.4: 2014	Compliance as per Subpart B & Subpart C FCC ID: S5751490045 BLE FCC ID: 2APD9-RSL10SIP
IC	ICES-003 Issue 7: 2020 ICES-Gen Issue 1: 2018+A1: 2021 RSS-247 Issue 2 Equipment Certification	Compliant for Wireless requirements IC ID: 573W-51490045 BLE IC ID: 23763-RSL10SIP

WARNING: Changes or modifications to this unit not expressly approved by the party responsible for compliance could void the user’s authority to operate the equipment.

NOTE:
This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation.
This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio reception, which can be determined by turning the equipment off and on,

the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Consult the dealer or an experienced radio technician for help.

NOTE:
This device contains licence-exempt transmitter(s)/receiver(s) that comply with Innovation, Science and Economic Development Canada's licence-exempt RSS(s). Operation is subject to the following two conditions:

* This device may not cause interference.
* This device must accept any interference, including interference that may cause undesired operation of the device.

L'émetteur/récepteur exempt de licence contenu dans le présent appareil est conforme aux CNR d'Innovation, Sciences et Développement économique Canada applicables aux appareils radio exempts de licence. L'exploitation est autorisée aux deux conditions suivantes:

* L'appareil ne doit pas produire de brouillage.
* L'appareil doit accepter tout brouillage radioélectrique subi, même si le brouillage est susceptible d'en compromettre le fonctionnement.

CAUTION: To maintain compliance with the FCC's RF exposure guidelines, place the unit at least 20cm from nearby persons.

Wireless Certifications & Approvals

Honeywell Versatilis Transmitter has LoRaWAN & BLE Wireless communication technologies. Required certifications and approvals have been attained for this product.

The below tables gives the summary on the same:

Table 11-5: Wireless certification and approvals

Certification	Standards/ Specification	Approval
LoRaWAN	LoRaWAN 1.0.4	End Device certification Requirements for all regions: Version 1.4
Bluetooth Low Energy (BLE)	Bluetooth Specifications	Bluetooth SIG Listed

NOTICES

Trademarks

Microsoft is a registered trademark or trademark of Microsoft Corporation in the United States and/or other countries.

Trademarks that appear in this document are used only to the benefit of the trademark owner, with no intention of trademark infringement.

Other trademarks

Other brands or trademarks are trademarks of their respective owners.

Trademarks that appear in this document are used only to the benefit of the trademark owner, with no intention of trademark infringement.

Third-party licenses

This product may contain or be derived from materials, including software, of third parties. The third party materials may be subject to licenses, notices, restrictions and obligations imposed by the licensor. The licenses, notices, restrictions and obligations, if any, may be found in the materials accompanying the product, in the documents or files accompanying such third party materials, in a file named third_party_licenses on the media containing the product.

Documentation feedback

You can find the most up-to-date documents in the Support section of the Honeywell Process Solutions website at: https://process.honeywell.com

If you have comments about Honeywell Process Solutions documentation, send your feedback to: hpsdocs@honeywell.com

Use this email address to provide feedback, or to report errors and omissions in the documentation. For immediate help with a technical problem, contact HPS Technical Support through your local Customer Contact Center, or by raising a support request on the Honeywell Process Solutions Support website.

How to report a security vulnerability

For the purpose of submission, a security vulnerability is defined as a software defect or weakness that can be exploited to reduce the operational or security capabilities of the software.

Honeywell investigates all reports of security vulnerabilities affecting Honeywell products and services.

To report a potential security vulnerability against any Honeywell product, please follow the instructions at:

https://www.honeywell.com/us/en/product-security.

Support

For support, contact your local Honeywell Process Solutions Customer Contact Center (CCC). To find your local CCC visit the website, https://process.honeywell.com/us/en/contact-us.

Training classes

Honeywell holds technical training classes that are taught by process control systems experts. For more information about these classes, contact your Honeywell representative, or see http://www.automationcollege.com.

Factory Information

Honeywell International (India) Pvt. Ltd., Plot No. 2, Gat No. 181, Village Fulgaon, Tal-Haveli, Pune, Maharashtra, 412216, India.